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Fundamentals of Data Science

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October 4, 2026

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Introduction

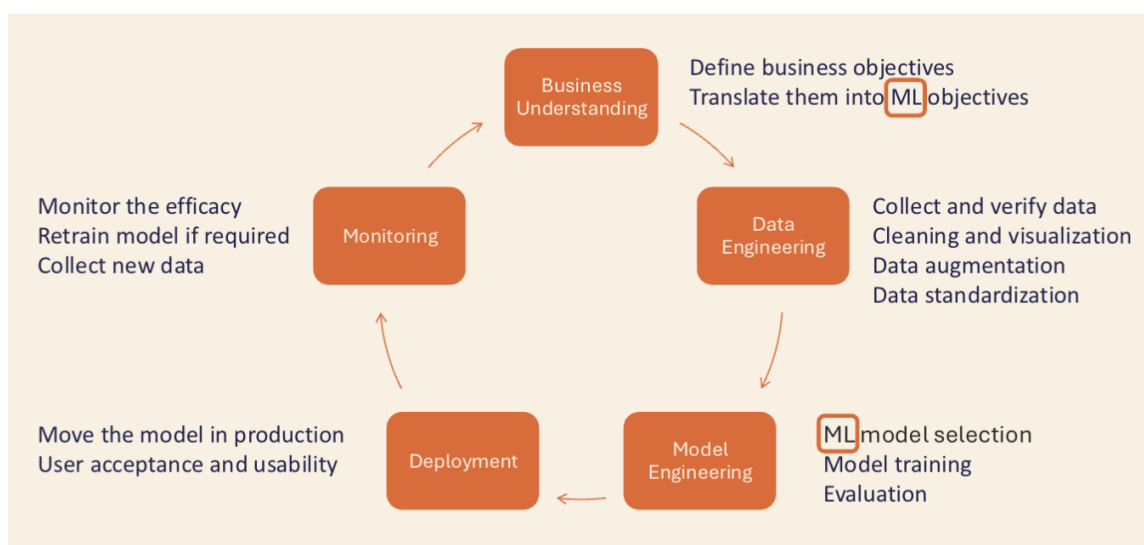
1.1 What is Data Science?

Definition 1.1: Data Science

Data Science is an interdisciplinary field that extracts knowledge from data to support decisions. It combines statistics, mathematics, computer science, and domain expertise to collect, clean, manage, and analyse data.

1.2 The Data Science lifecycle

A project moves through business understanding, data engineering, model engineering, deployment, and monitoring. This cycle can repeat when new data or retraining is needed.



The Data Science lifecycle. Source: Spinelli, Lecture 00.

1.3 Artificial Intelligence and Machine Learning

Artificial Intelligence (AI) studies how to build systems that perform tasks requiring abilities such as reasoning, learning, and problem solving.

Definition 1.2: Machine Learning

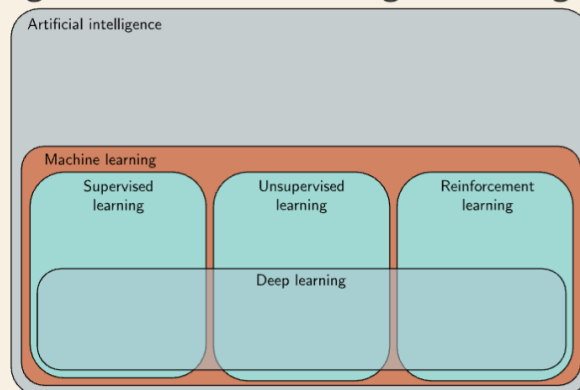
Machine Learning (ML) studies algorithms whose performance P on a task T improves through experience E . A learning problem therefore specifies the task, the performance measure, and the available experience.

The main learning settings are:

- **Supervised learning:** learn from inputs paired with known desired outputs to predict the output for new inputs.
- **Unsupervised learning:** discover structure in data without given target outputs.
- **Reinforcement learning:** learn how to act through interaction with an environment and feedback in the form of rewards.

What is ML (Machine Learning)

Artificial Intelligence is the field of computer science dedicated to creating agents capable of performing tasks that typically require human intelligence such as learning reasoning and problem solving.



AI, ML, and learning settings. Source: Spinelli, Lecture 00.

Deep learning uses neural networks with multiple layers and can be applied in each of these settings.

1.4 Supervised models and prediction tasks

Definition 1.3: Supervised learning model

A supervised learning model is a function $f : \mathcal{X} \rightarrow \mathcal{Y}$ that maps an input x to a predicted output $\hat{y} = f(x)$. Here, $\mathcal{X}$ is the input space and $\mathcal{Y}$ is the output space. Training means choosing or adjusting f using examples of inputs paired with their known outputs.

Definition 1.4: Regression

Regression is a supervised learning task whose target is a real-valued number or vector. For example, predicting a person's height from their age is a regression task.

Definition 1.5: Classification

Classification is a supervised learning task whose target is a category from a finite set of classes. For example, predicting whether a flower is setosa, versicolor, or virginica is a classification task.

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Data Representation

Start of lecture 2 October 2026 (Spinelli)

2.1 The Iris dataset

The Iris dataset contains 150 flowers, divided into three species: 50 setosa, 50 versicolor, and 50 virginica. The goal is to predict a flower's species from its measurements.

Example 1: Iris as a supervised classification task

Each **observation** is one flower, described by four **features**, all measured in centimetres:

- sepal length and sepal width;
- petal length and petal width.

The **target** is the flower's species. This is supervised learning because the species is known for the training examples, and classification because the target is a category.